

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – December 2011

PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 07.12.11

Time: 10.00 a.m to 10.20 a.m

Maximum Marks: 10

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(1 × 10 = 10)

1. The refractive index of the cladding material compare to that of the core in optical fiber is
 - a. Less than 1
 - b. Greater than 1
 - c. Equal to 1
 - d. Not any of above
2. The supersonic sound wave has velocity.....
 - a. > 330 m/s
 - b. < 330cm/s
 - c. 330 m/s
 - d. 332.8m/s
3. What is the Newton's second law?
 - a. $E=hf$
 - b. $F=m \cdot dv/dt$
 - c. $v = \lambda f$
 - d. $F= m \cdot de/dt$
4. What is the unit of Resistivity?
 - a. ohm-meter
 - b. ohm
 - c. moh
 - d. unit less
5. What is the First law of thermodynamics?
 - a. $dQ=dU/PdV$
 - b. $dQ=dU-PdV$
 - c. $dQ=dU+PdV$
 - d. $dQ = dU \cdot PdV$

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY
CHARUSAT CAMPUS CHANGA

Examination Semester _____

Subject _____ No. _____

Candidate's Seat No. _____

Section _____

Date _____

Sr. Supervisor's
Sign.

6. If frequency of emitted X-ray $\nu = 2 \times 10^{17}$ Hz. Calculate the X-ray wavelength.
- 1×10^{-8} m
 - 1.5×10^{-9} m
 - 2×10^7 m
 - 1.5×10^9 m
7. The ripple factor in full wave rectifier is 0.48 and efficiency is $\sim 81\%$.
- True
 - False
8. A person produces sound and this sound is reflected back from an obstacle at a distance x . The velocity of sound $v=360$ m/s. the time taken by the sound to travel to the obstacle and back is $t=0.1$. Calculate the distance between the person and obstacle x .
- 18 m
 - 24 m
 - 30 m
 - 20 m
9. What is the unit of Numerical Aperture?
- Degree
 - eV
 - minute
 - unit less
10. The unit of Peltier co-efficient is given by.....
- Amp
 - Ohm
 - Volt
 - No unit

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PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 07.12.11

Time: 10.20 a.m to 1.00 p.m

Maximum Marks: 60

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2)

(1× 10= 10)

- a) What is Doppler Effect? Derive the equation for the frequency experienced by the listener under Doppler condition.
- b) Obtain relation explaining effect of temperature and sound wave.
- c) Give at least three chemical application of sound wave.

OR

Give a detail explanation of X-ray.

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Explain production of X-ray.
2. Explain Laplace correction for sound velocity.
3. What is diathermy? Explain in brief shortwave diathermy.
4. Explain Piezo electric effect in brief.
5. Give the difference between stimulated and spontaneous emission.

SECTION –II

Q.3. Long Question (1 out of 2)

(1× 10= 10)

Explain basic principle of fiber optic. Derive formula of numerical aperture and acceptance angle.

OR

Explain electromagnetic induction, self-inductance and mutual inductance.

Q.4. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. A laser source emits a light of wavelength $0.621\mu\text{m}$ and has an output of 35mW. Calculate how many photons are emitted per minute by this laser source.
2. Write note on: (i) Seebeck effect, (ii) Peltier effect
3. Explain different types of magnetic materials.
4. Draw the diagram of transistor as amplifier and explain it.
5. Explain Zener diode as a voltage regulator.